

SENTIMENT ANALYSIS FOR MENTAL HEALTH REPORT

DEEP LEARNING PROJECT

PROJECT REPORT

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1. INTRODUCTION

Mental health has become a significant concern in modern society, especially with the increasing use of social media platforms and online communication. People often express their emotions, stress levels, and psychological conditions through written messages, posts, and comments. Analyzing such textual data can provide valuable insights into an individual's mental state and emotional well-being.

Traditional machine learning techniques for text classification rely on feature extraction methods such as Bag of Words and TF-IDF. Although these methods can classify text effectively, they often fail to capture contextual relationships between words in a sentence.

Recent advancements in Deep Learning have introduced powerful models such as Recurrent Neural Networks (RNN). Long Short-Term Memory (LSTM) networks are particularly effective in learning patterns from sequential textual data.

This project focuses on developing an LSTM-based RNN model to classify mental health related statements into different sentiment categories. The trained model is integrated into a Streamlit web application that allows users to enter text and obtain real-time sentiment predictions.

2. ABSTRACT

This project focuses on analyzing mental health related textual statements using a Long Short-Term Memory (LSTM) based Recurrent Neural Network model. Mental health sentiment analysis helps identify emotional patterns in textual data and provides insights into psychological conditions expressed through written statements.

The objective of this project is to develop a deep learning model capable of classifying mental health related text into different sentiment categories. The dataset consists of user statements that represent different emotional conditions.

The project involves several stages including text preprocessing, cleaning, tokenization, sequence padding, and deep learning model development. An LSTM based RNN model was trained to learn patterns from sequential text data.

Additionally, a Streamlit web application was developed to allow users to input text statements and obtain real-time sentiment predictions. The trained model demonstrates good performance in classifying mental health related sentiments.

3. OBJECTIVES

- Build an LSTM-based text classification model for mental health sentiment analysis
- Perform text preprocessing and data cleaning
- Convert textual data into numerical sequences using tokenization
- Train and evaluate a deep learning model using the dataset
- Analyze model performance using evaluation metrics
- Develop a Streamlit web application for real-time sentiment prediction

4. METHODOLOGY

Step 1: Data Collection

The dataset used in this project contains textual statements related to mental health conditions. The dataset includes statements that reflect various emotional states and psychological conditions.

The dataset was obtained directly from Kaggle.

Step 2: Data Preprocessing

- Conversion of text to lowercase
- Removal of special characters and punctuation
- Removal of stopwords
- Tokenization using Keras Tokenizer
- Conversion of text into numerical sequences
- Padding of sequences to maintain equal input length

Step 3: Train-Test Split

Dataset split into:

- 80% Training Data
- 20% Testing Data

Step 4: Model Building

A Long Short-Term Memory (LSTM) based Recurrent Neural Network model was developed.

Model architecture includes:

- Embedding Layer
- LSTM Layer
- Dense Layer
- Dropout Layer
- Output Layer

Step 5: Model Evaluation

The model was evaluated using:

- Accuracy Score
- Classification Report
- Confusion Matrix
- Training and Validation Accuracy

Step 6: Deployment

A Streamlit web application was developed to allow users to enter text statements and obtain real-time mental health sentiment predictions.

5. DATASET OVERVIEW

Source: – Mental Health Sentiment Dataset(Kaggle)

Features:

- Statement → User text related to emotional condition
- Status → Target variable representing mental health sentiment
- The dataset consists of textual statements that describe the emotional state of individuals. These statements are labeled according to their mental health status.

6. DATA ANALYSIS AND EDA VISUALS

Basic dataset exploration and analysis were performed to understand the structure and characteristics of the dataset.

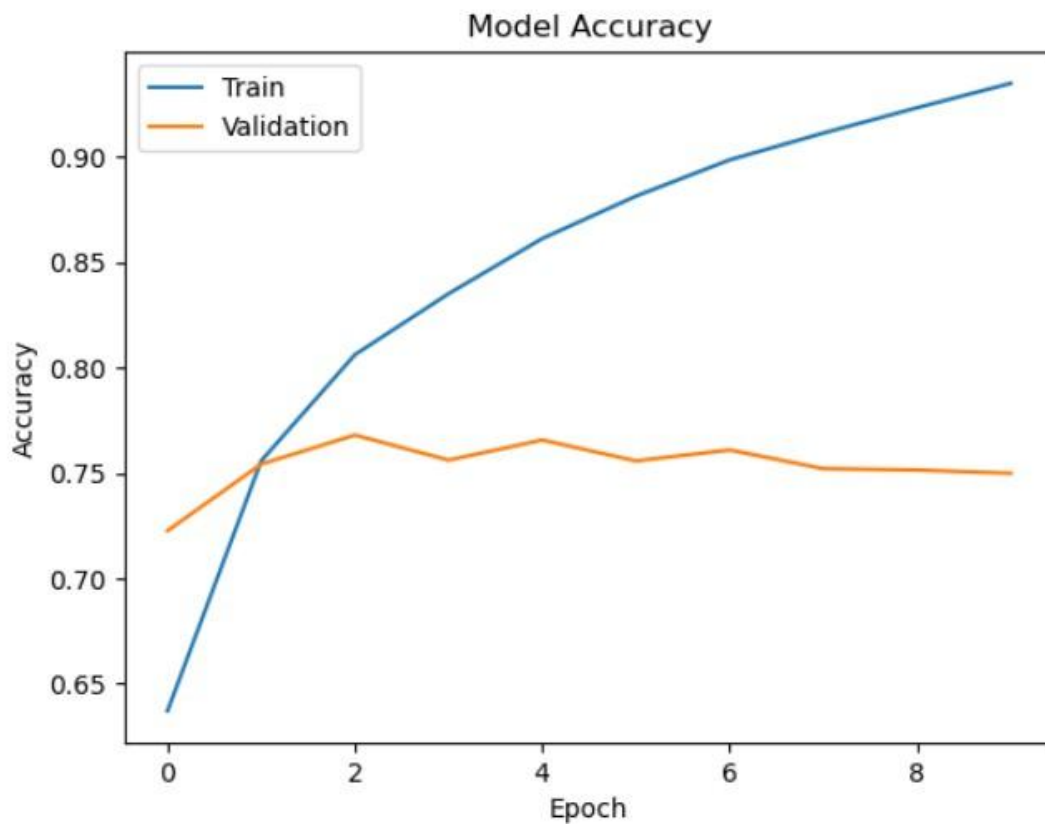
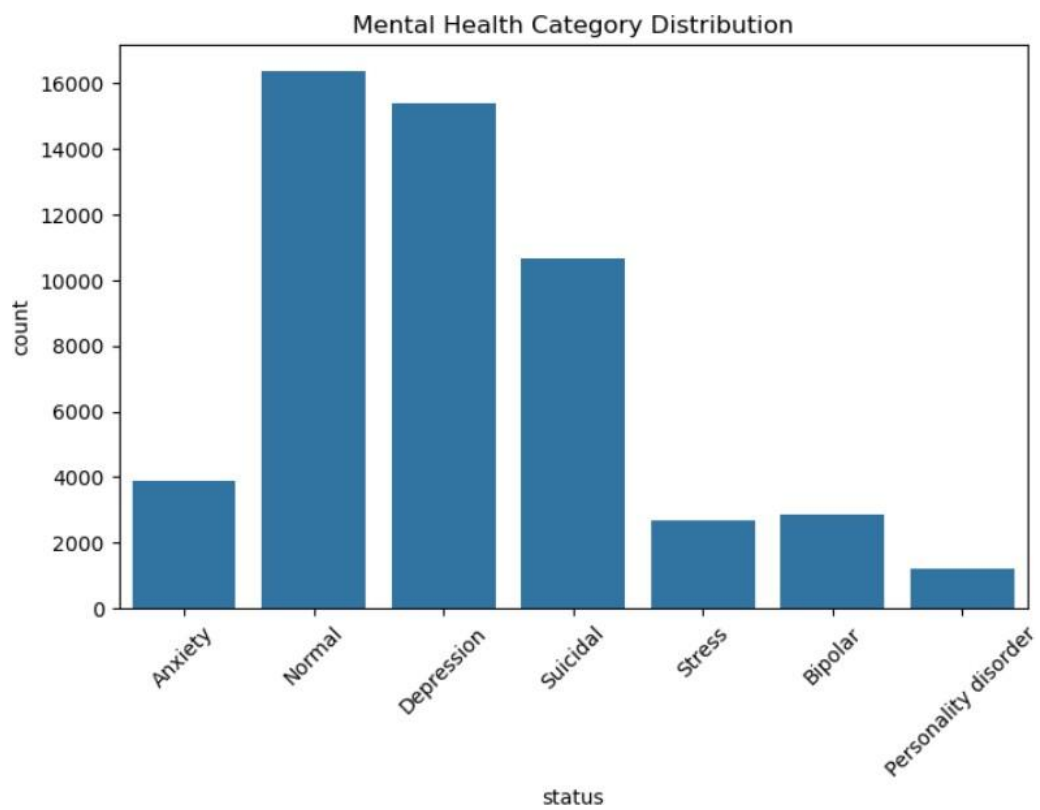
The following steps were carried out:

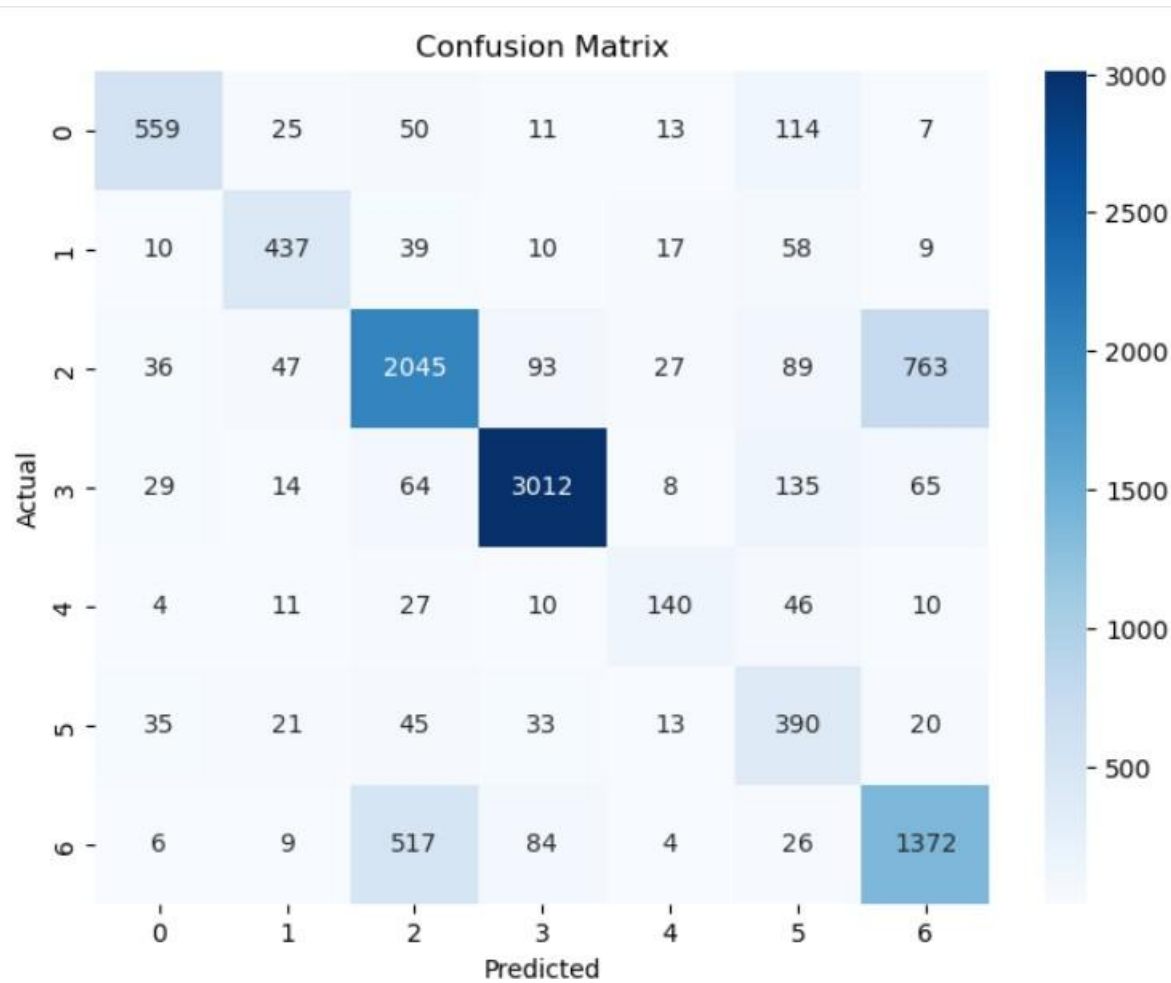
- Checked dataset shape and structure
- Verified dataset columns and information
- Converted text column into string format
- Checked for missing values
- Analyzed distribution of sentiment classes

Visualizations:

Basic exploratory data analysis was performed to understand the structure and distribution of the dataset. Visualization techniques were used to analyze the distribution of mental health categories and to better understand the patterns present in the dataset.

The following visualizations were generated during the analysis phase.





7. MODEL DEVELOPMENT & RESULTS

Model Applied:

LSTM-based Recurrent Neural Network (RNN) model

- **Model Architecture:**

- ~ Embedding Layer (Word To Vector Representation)
- ~ LSTM Layer (Capture Sequential relationships between words)
- ~ Dense Layer (Extracts higher level features)
- ~ Dropout Layer (Reduces overfitting)
- ~ Output Layer (predicts sentiment category using Softmax activation)

- **Compilation:**

- ~ Optimizer → Adam
- ~ Loss Function → Sparse Categorical Crossentropy
- ~ Evaluation Metric → Accuracy

- **Training Configuration:**

- ~ Epochs → 10
- ~ Batch Size → 64

Performance:

- The trained model achieved good classification accuracy in predicting mental health sentiment categories from textual statements. Training and validation accuracy curves indicate that the model successfully learned patterns from the dataset.
- The classification report provides detailed metrics such as precision, recall, and F1-score for each sentiment category.

8. STREAMLIT WEB APP DEPLOYMENT



Sentimental analysis for mental health

Analyze emotions from text using RNN (LSTM)

Enter your text here

I constantly feel nervous and worried about things that might go wrong.

 Predict Sentiment

Predicted Mental State: Anxiety 😟



Sentimental analysis for mental health

Analyze emotions from text using RNN (LSTM)

Enter your text here

I feel empty and sometimes think about ending my life

 Predict Sentiment

Predicted Mental State: Suicidal 🚨



Sentimental analysis for mental health

Analyze emotions from text using RNN (LSTM)

Enter your text here

There is too much pressure in my life and I feel mentally exhausted all the time

 Predict Sentiment

Predicted Mental State: Depression 😞



Sentimental analysis for mental health

Analyze emotions from text using RNN (LSTM)

Enter your text here

I had a good day

 Predict Sentiment

Predicted Mental State: Normal 😊

9. CONCLUSION & FUTURE SCOPE

Conclusion:

The LSTM-based Recurrent Neural Network model successfully classifies mental health related textual statements into different sentiment categories. The model demonstrates the effectiveness of deep learning techniques in analyzing textual data related to emotional and psychological states.

Integration with a Streamlit web application allows users to interact with the model and obtain real-time sentiment predictions.

Future Scope:

- The model accuracy can be improved by training with a larger and more diverse mental health dataset containing a wider range of emotional expressions.
- The system can be integrated with social media platforms to analyze real-time user posts and detect mental health related sentiments automatically.
- A real-time mental health monitoring system can be developed that provides early warning signals for individuals experiencing severe emotional distress